BME 5910 Homework 6

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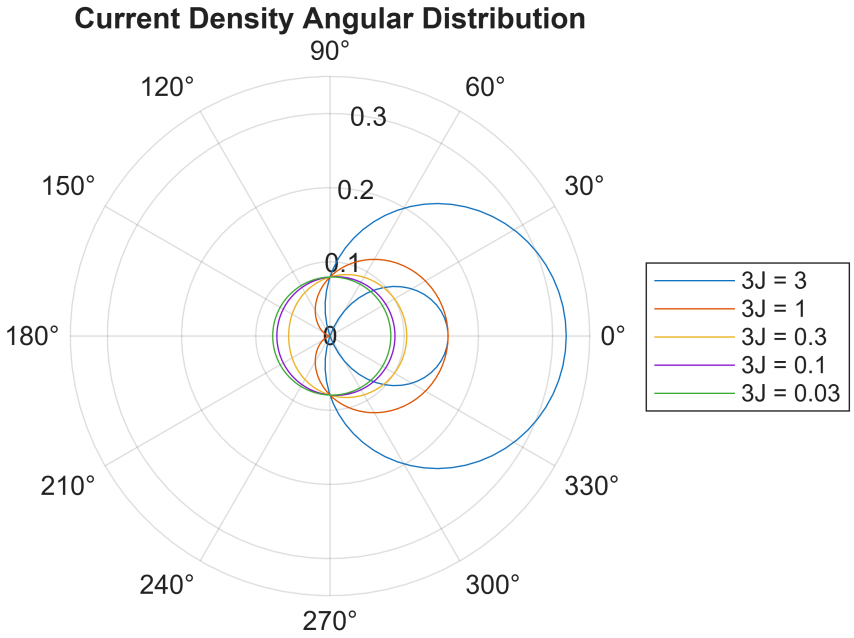
10/13/2026

Problem 5.9

```
clear, clf
% Givens:

Phi = 1;
J_threemag = [3,1,0.3,0.1,0.03];
alpha = linspace(0,2*pi,100);

% Plotting
figure
for j_3 = J_threemag
    L = 1/4/pi*(1+j_3*cos(alpha));
    polarplot(alpha, L, 'DisplayName', sprintf('3J = %.2g', j_3));
    hold on
end
legend
title("Current Density Angular Distribution")
```



Problem 5.10

```
% Givens

theta = linspace(0,2*pi,100);
```

```
g_values = [0,0.1,0.5,0.9,0.99];

% Plotting
tiledlayout(2, 3, 'Padding', 'compact', 'TileSpacing', 'compact');
for g = g_values
    P = 1/4/pi*(1-g^2)./(1+g^2-2*g*cos(theta)).^1.5;
    nexttile
    polarplot(theta,P);
    rlim([0,max(P)*1.2])
    title(sprintf('g = %.2g', g))
end
sgtitle("Henyey-Greenstein Phase Function")
```

